

DESIGN OF MULTI-LABORATORY EXPERIMENT TO ASSESS INTER-LAB RELIABILITY

Statistical Considerations

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Primary Objective

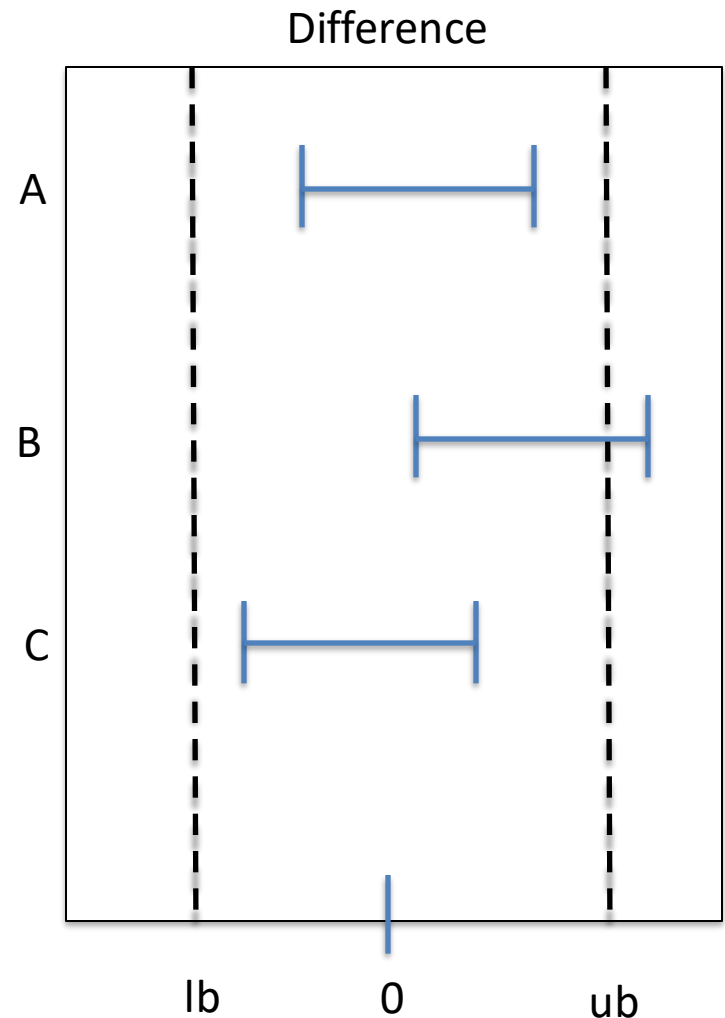
- Determine the inter-lab variability in a multi-site, randomized and blinded chronic pain treatment study in mice and rats
- Hypothesis: Variance in behavioural outcomes before and after treatment across different laboratories with experience in pain pharmacology will be modest

General Statistical Approaches To Consider

- Superiority testing
 - Site as fixed effect
- Variance components
 - Random effects (Generalizability Theory)
 - Reliability analysis
- Frequentist replication
 - Proportion replicating
 - Meta-analysis (Heterogeneity, I^2)
- Latent Trait Modeling
 - Latent Growth Curves
- Equivalence testing
 - Formal inference equivalence region

Equivalence Testing

- Hypothesis testing used to generate a formal inference to evaluate if groups are equivalent on an outcome
 - Null hypothesis and alternative hypothesis are switched compared with superiority testing
- Must define an equivalence region (upper and lower bounds) used to define uninteresting variation



Multiple Group Equivalence Testing

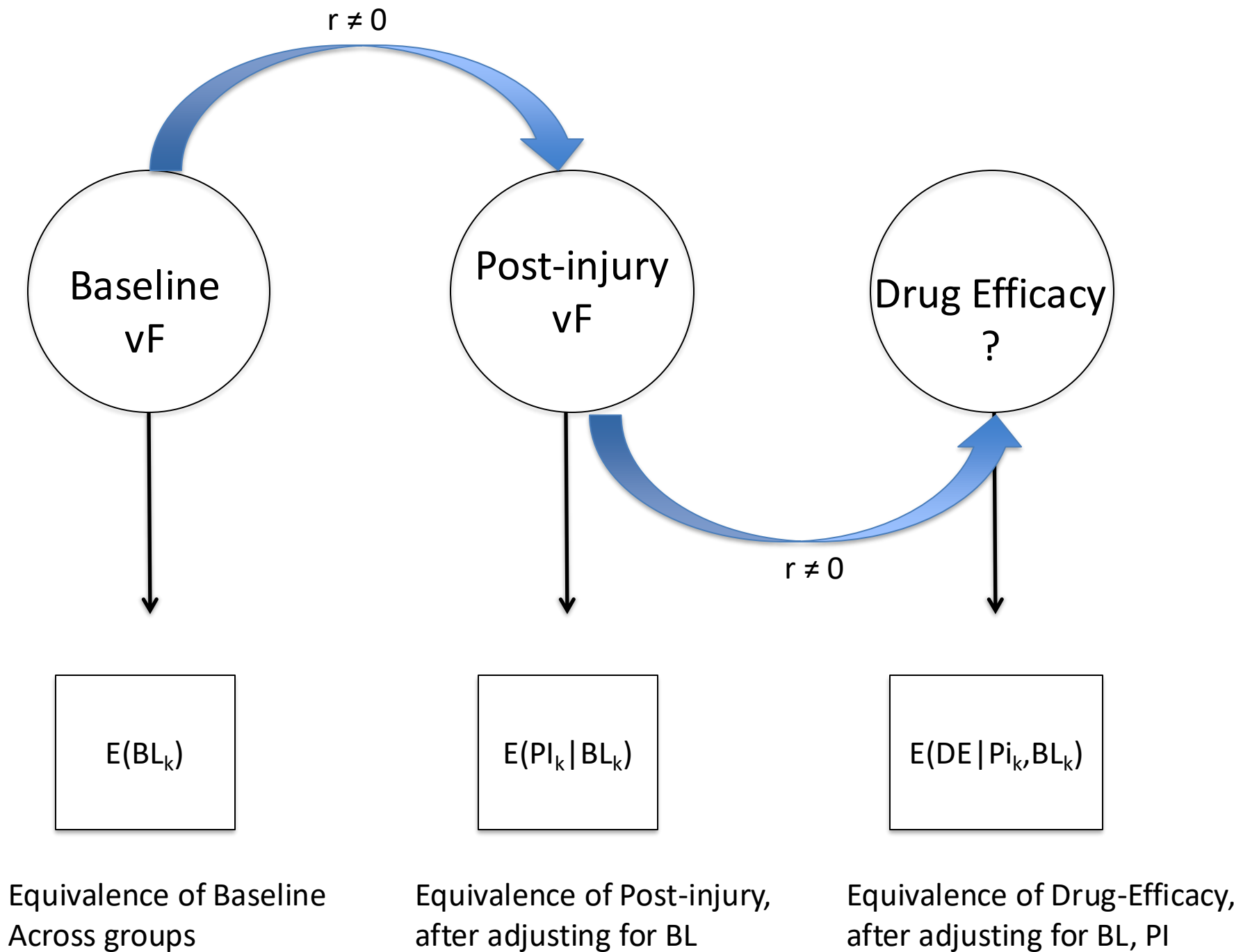
- Two One-Sided T-Tests (TOST)
 - Conducted for k groups, adjusted alpha ($k-1$)
- F-Test for Equivalence (Wellek, 2010)
 - ANOVA on Euclidian distance
- Range Test
 - Examine greatest difference for equivalence
- Region of Practical Equivalence (ROPE)
 - Bayesian using HDI in equivalence region

F-Test for Equivalence (Wellek, 2010)

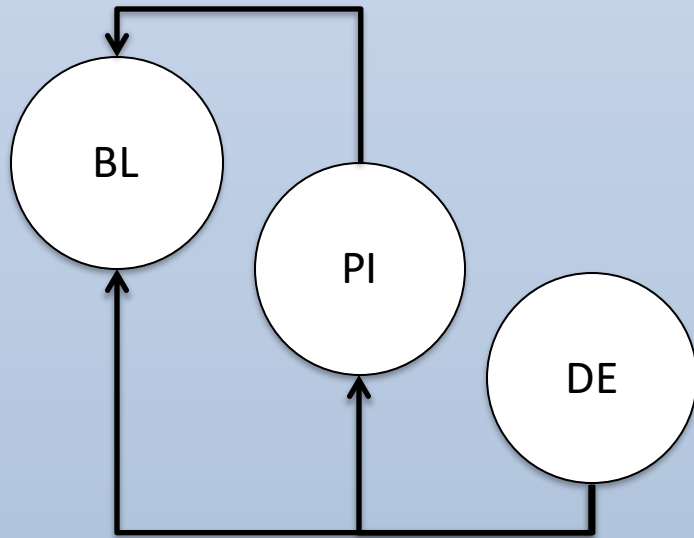
- The distance of each site (group) is summed as the squared Euclidian distance from the grand mean
- Noncentral version of one-way ANOVA

$$\psi^2 = \sum_{i=1}^k \frac{(\mu_i - \bar{\mu}_{.})^2}{\sigma^2}$$

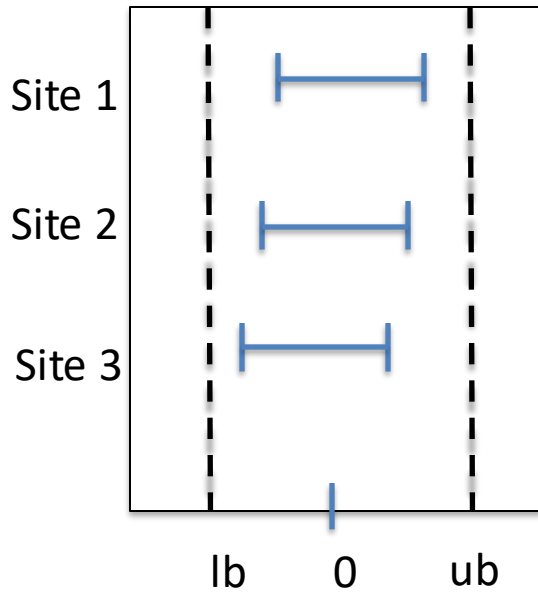
$$DR = \sqrt{\frac{\theta^2 + 2\sigma^2}{2\sigma^2}},$$



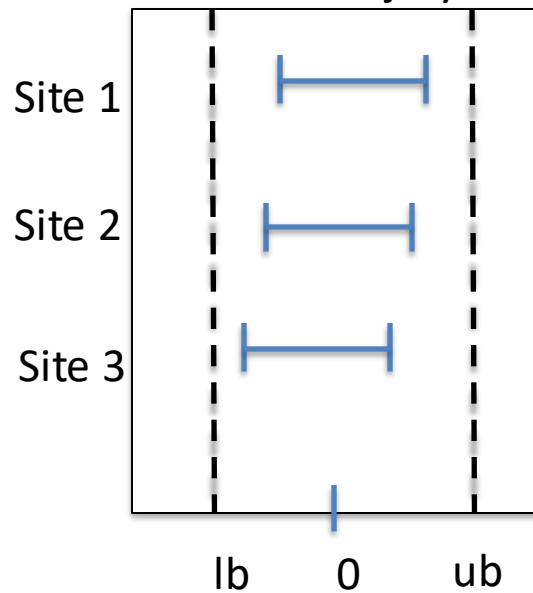
Primary: Multigroup Equivalence Testing



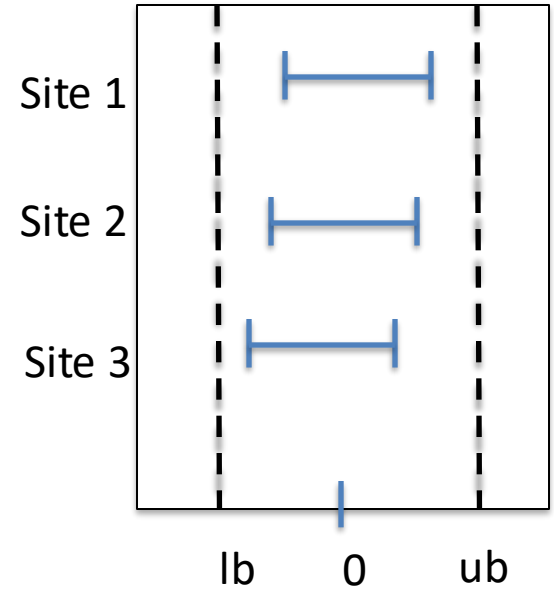
Baseline



Post-Injury



Drug Effect



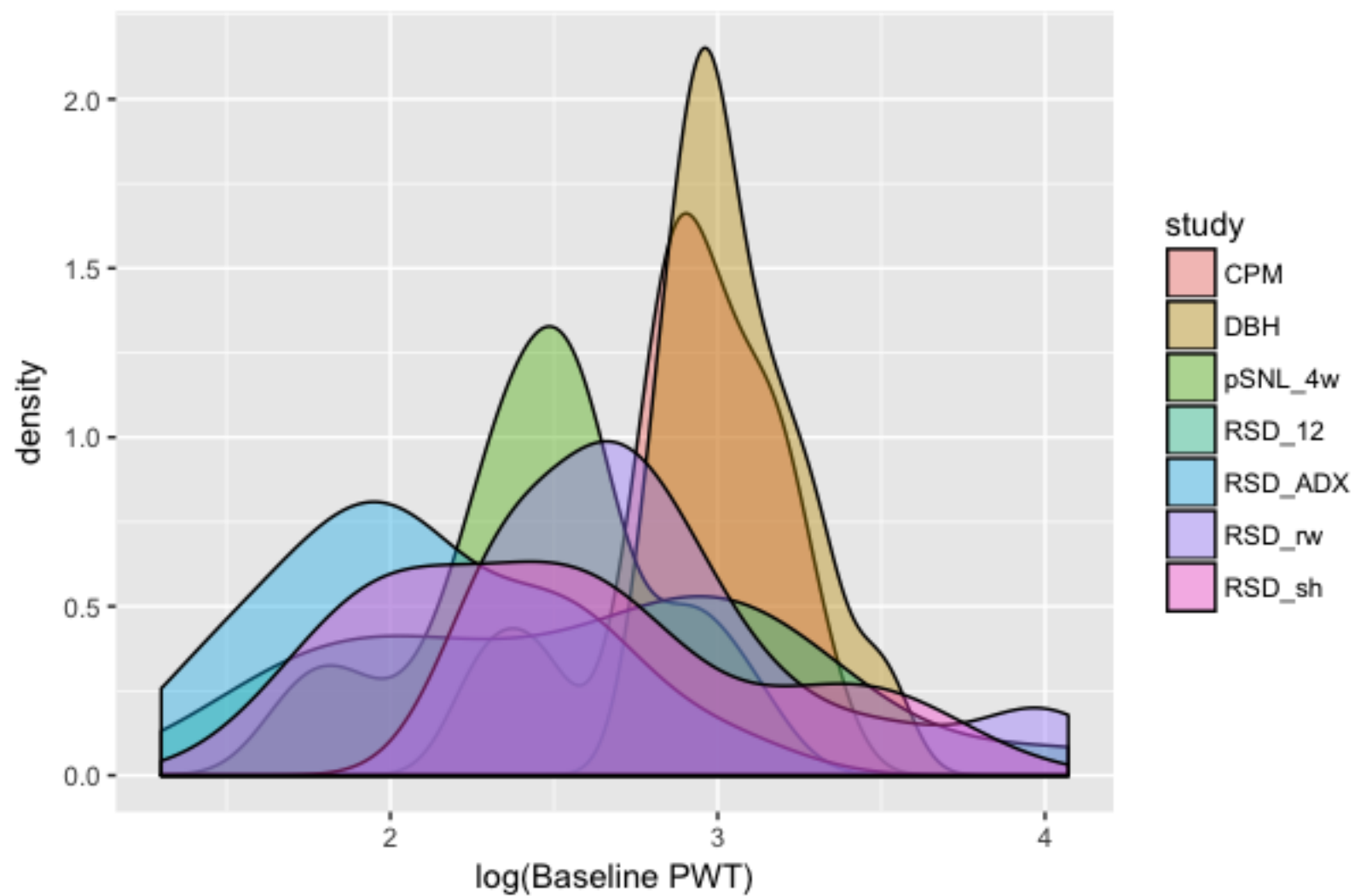
Proposal

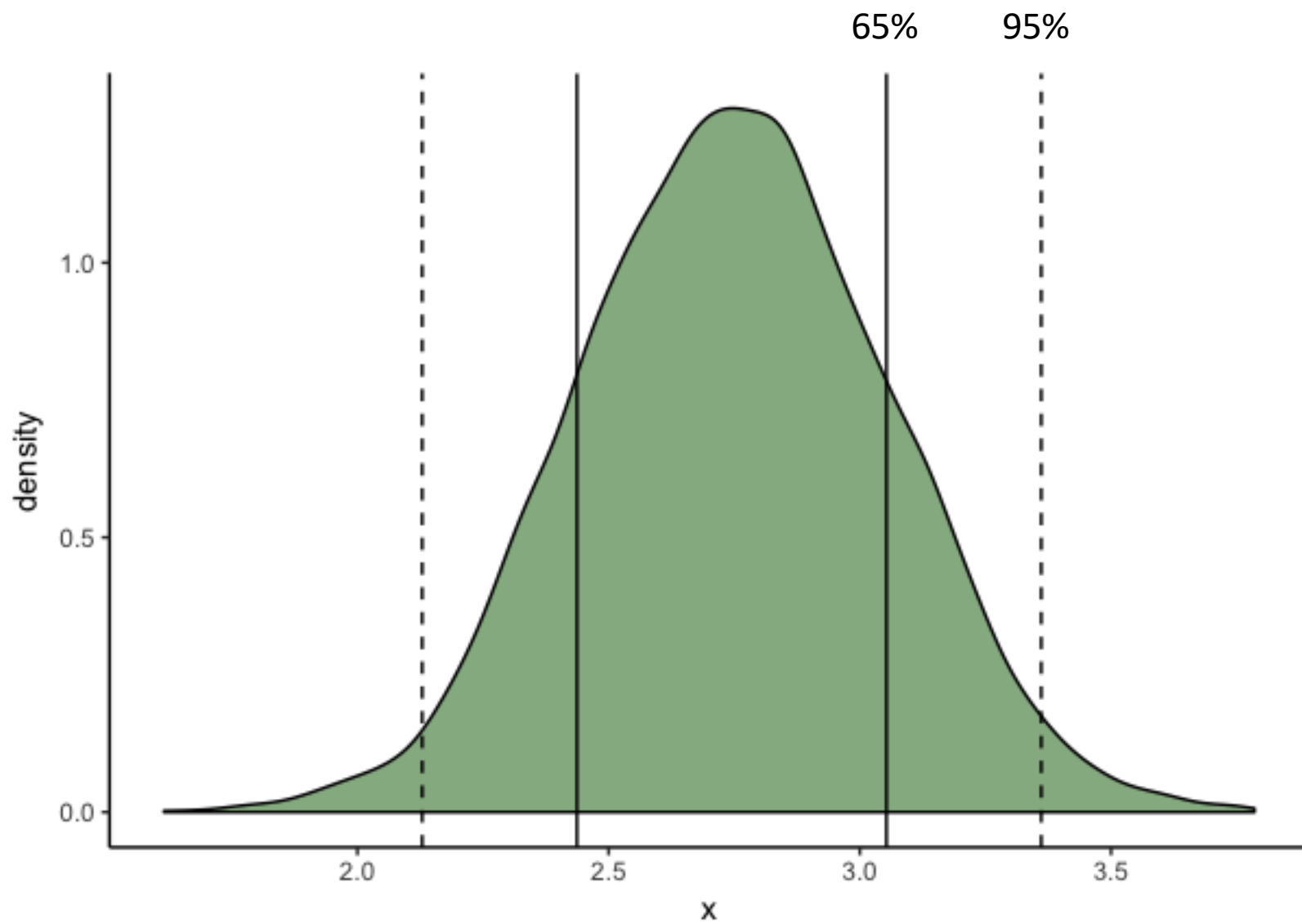
- (Primary) Conduct multi-group equivalence testing on six families of inferences
 - By animal type (Rat, Mice)
 - By (adjusted) outcome (BL, PI, DE)
- (Secondary) If desired, follow-up non-equivalence findings with post hoc TOSTs (Adjusted for FWER)
- (Secondary) Estimate within-site fixed effects models to examine the findings and replication probabilities
 - Aggregate effects using meta-analyses
- (Secondary) Conduct variance components models to partition variance into measurement facets (i.e., prediction model)
 - Estimate a partitioned ICC for each effect

Unresolved Issues

- Equivalence Region?
- Alpha error rate for primary analysis (i.e., to adjust?)
- Missing Data
- Outliers
- Additional covariate adjustment

Extra Slides





Eisenach Lab Data: Baseline

- $N = 7$ studies
- Distribution of PWT
- $\log(\text{PWT})$
 - Mean: 2.745
 - SD: .308
 - Within-study SD: 1.923

Power

Equivalence Bounds (SD)	Sample Size (per group)
0.25	274
0.50	69
0.75	30
1.0	17
1.25	11
1.50	8

